

PHC Pulse

A heartbeat for every health centre in India.

THE PROBLEM

India's PHC network runs blind on stock, beds, and staff.

Public healthcare systems face persistent supply chain vulnerabilities. Without real-time visibility into medicines, patient footfall, and resource utilisation across vast Primary Health Centre networks, stock-outs happen — and the system can't respond fast when it matters most.



Invisible Stock-Outs

Medicine shortages are discovered when a patient is turned away, not before.



No Real-Time Footfall Data

Bed availability and patient load are tracked on paper registers, weeks behind reality.



No Redistribution Mechanism

Nearby PHCs may hold surplus stock with no system to move it before a shortage hits.

OUR SOLUTION

PHC Pulse gives every Primary Health Centre a one-minute-a-day voice report — and turns it into national-scale forecasting and automatic redistribution.

ASHA and PHC staff log stock, beds, and attendance by voice in their own language. Vertex AI forecasts shortages weeks ahead and recommends where to move surplus stock before it runs out.



Voice-First Reporting

Under 60 seconds a day, in the health worker's own language — no new hardware, no new habit.



Demand Forecasting

Per-PHC, per-drug forecasts flag stock-outs 2-3 weeks before they happen.



Redistribution Recommender

A prescriptive model suggests moving surplus stock across districts under transport-cost constraints.

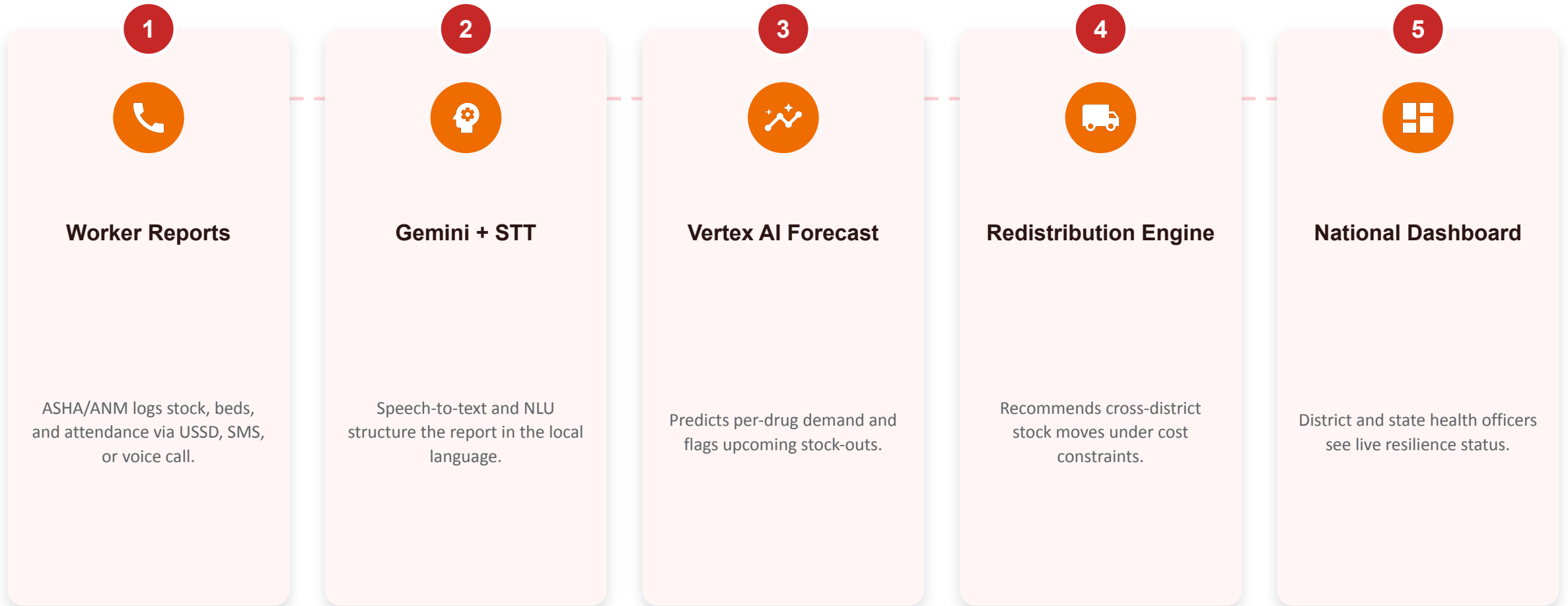


Outbreak Anomaly Detection

Unusual footfall spikes are flagged automatically as early emergency signals.

HOW IT WORKS

From a health worker's voice note to a same-day resupply.



Built for the lowest-bandwidth environment first.



Gemini + Speech-to-Text

Voice-first stock and attendance reporting in local language for low-literacy health workers.



Vertex AI AutoML

Per-PHC, per-drug demand forecasting using historical usage and seasonal/outbreak signals.



Vertex AI Optimization

Prescriptive cross-district redistribution recommendations under transport-cost constraints.



BigQuery

National PHC data warehouse for stock, beds, and staff attendance with row-level state access control.



Dialogflow (USSD/IVR)

Works on basic feature phones — no smartphone or app required for reporting.



Vertex AI Anomaly Detection

Flags unusual footfall spikes as early outbreak signals for rapid response.

A prescriptive system, not just another dashboard.



Prescriptive, not just descriptive

PHC Pulse doesn't just show a stock-out coming — it recommends exactly which PHC should ship what, and when.



Voice-first for feature phones

Reaches PHCs with poor connectivity and no smartphones through USSD and IVR, not a mobile app.



HMIS-compatible by design

Built to integrate with existing Health Management Information Systems, not replace them.

WHO IT SERVES

From the front line to the health ministry.



ASHA & PHC Staff

Near-zero-effort daily reporting — under 60 seconds, no new hardware, in their own language.



District Health Officers

A real-time dashboard instead of a monthly paper report — see problems as they form.



State & National Health Ministry

Cross-state redistribution and emergency response coordination from one live view.

Three screens, from a phone call to a resupply order.



Voice Reporting Flow

- "How many paracetamol strips remain?" — answered by voice
- Bed count and staff attendance logged in the same call
- Confirms totals back in the worker's language



District Officer Dashboard

- Live stock, bed, and attendance status by PHC
- Early-warning flags for predicted stock-outs
- Outbreak anomaly alerts by footfall spike



Redistribution Alert

- "PHC-14 has surplus paracetamol; PHC-22 needs it in 9 days."
- One-tap approval for cross-district transfer
- Tracks delivery status to resolution

WHY IT'S DEPLOYABLE

Designed for the PHCs that need it most.



Works over USSD and basic voice calls

Reaches PHCs with poor connectivity or no smartphones at all.



Under 60 seconds a day

Minimal workflow change for already-overloaded health workers.



HMIS-integration-first

Built on the existing Health Management Information System data model, not a replacement system.



Generalizes beyond medicine

The same redistribution model extends to blood banks, oxygen, and ambulances.

One engine, onboarded state by state.

Phase 1 · Pilot

100 PHCs across 2 districts in one state, validating the voice-reporting workflow.

Phase 3 · Cross-State Redistribution

Enable the optimization engine to recommend transfers across state boundaries.

Phase 2 · State Rollout

Connect to the state's HMIS data pipeline and add a local language pack.

Phase 4 · Resource Generalization

Extend the same engine to blood banks, oxygen supply, and ambulance networks.

IMPACT & THE ASK

80%+

of stock-out cases target flagged 2+ weeks ahead,
versus 0% reactive today

100 PHCs

in the pilot, across 2 districts of one state

Same-day

target redistribution decision time, down from weeks of
manual coordination

THE ASK

We're looking for a state health department pilot partner, access to anonymized historical HMIS data, and Vertex AI credits to train the forecasting and redistribution models.

PHC Pulse — a heartbeat for every health centre in India.